THOUGHT AND ACTION TENDENCIES AND EMOTION

- **Thought–action tendencies :** the way feelings guide behavior and information processing is through the urges that accompany them (Fredrickson, 1998)

Emotions and their associated thought–action tendencies *Twelve emotions and the urges they spark.* (After Fredrickson, 1998, 2002; Fredrickson & Branigan, 2001; Frijda, 1986; Lazarus, 1991b)

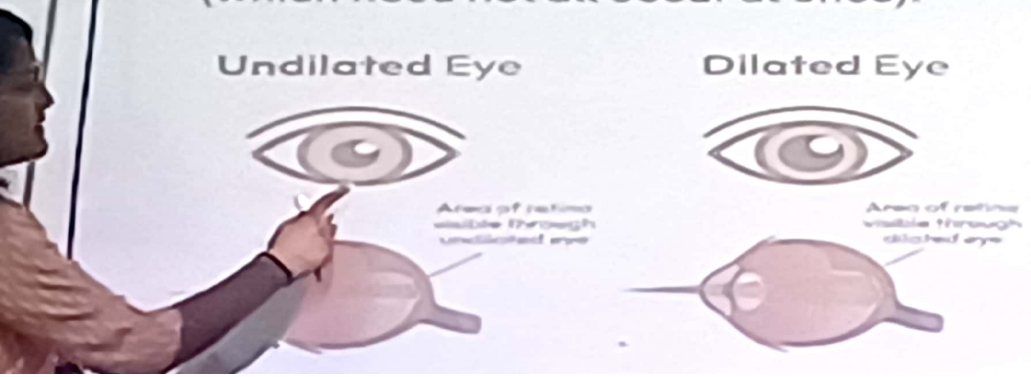
Emotion	*Thought–action tendency*
Anger	Attack
Fear	Escape
Disgust	Expel
Guilt	Make amends
Shame	Disappear
Sadness	Withdraw
Joy	Play
Interest	Explore
Contentment	Savor and integrate
Pride	Dream big
Gratitude	Be prosocial
Elevation	Become a better person

Remember Margaret Meade?

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Physiology of Emotions: BODILY CHANGES AND EMOTION

- Result from activation of the **sympathetic division** of the **autonomic nervous system**
- Sympathetic nervous system prepares the body for emergency action and is responsible for the following changes (which need not all occur at once):

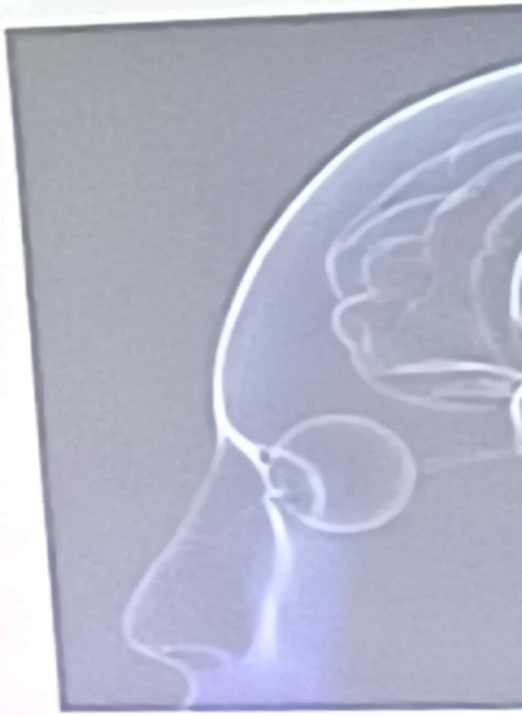


1. Blood pressure and heart rate increase.
2. Respiration becomes more rapid.
3. The pupils dilate.
4. Perspiration increases while secretion of saliva and mucus decreases.
5. Blood-sugar level increases to provide more energy.
6. The blood clots more quickly in case of wounds.
7. Blood is diverted from the stomach and intestines to the brain and skeletal muscles.
8. The hairs on the skin become erect, causing goose pimples.

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Appraisals in the brain

- Research on the brain circuits involved in emotion processes also supports the view that appraisals occur both consciously and unconsciously.
- One brain structure that plays key role within emotion circuits **is the amygdala**, a small, almond-shaped mass that is located in the lower brain and is known to register emotional reactions (Phelps, 2006).
- At one time, it was thought that the amygdala received all its inputs from the cortex and, hence, that those inputs always involved conscious appraisal.
- But research with rats uncovered connections between sensory channels and the amygdala that do not go through the cortex, and these direct connections may be the biological basis of unconscious appraisals (LeDoux & Phelps, 2000).
- The amygdala is capable of responding to an alarming situation before the cortex does, which suggests that sometimes we can experience an emotion before we know why.
- Although the initial research on the amygdala's role in automatic emotions was based on rats, the neural pathways involved appear similar in humans (Phelps, 2006).
- Brain imaging in humans has also demonstrated a key role for the amygdala within emotions



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Role of Amygdala

- Using the **backward masking technique**, fearful facial expressions were shown to participants for about 30 milliseconds and then masked by neutral expressions on the same faces.

Fearful - 30 ms
Neutral - Longer

- These data suggest that the amygdala monitors emotion-eliciting stimuli at an automatic, nonconscious level.

- Ted Bundy: antisocial personality disorder less activation in amygdala (Kiehl et al., 2001)



y = −6

Figure 11.3 Amygdala Activation During Unconscious Appraisals. This image shows a coronal slice of brain. Fearful and happy faces were shown to participants using a backward masking technique that prevented conscious awareness. Portrayed is the activation difference for masked fearful versus masked happy faces. The highlighted area represents greater activation in the amygdala for masked fearful faces. These findings suggest that the amygdala may play a role in the unconscious appraisal of emotionally-relevant stimuli. (From Whalen, Rauch, Etcoff, McInerney, Lee, & Jenike, (1998). 'Masked Presentations of Emotional Facial Expressions Modulate Amygdala Activity Without Explicit Knowledge', Journal of Neuroscience 18, 411–418.)

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Activities in the Brain and the ANS

- These activities of the autonomic nervous system are themselves triggered by activity in certain regions of the brain, including the
- **hypothalamus**
- **Amygdala**
- Impulses from these areas are transmitted to nuclei in the brain stem that control the functioning of the autonomic nervous system.

- The autonomic nervous system then acts directly on the muscles and internal organs to initiate some of the bodily changes described.
- It also acts indirectly by stimulating the adrenal hormones to produce other bodily changesAs the emotion subsides, the **parasympathetic nervous system** – the energy conserving system – takes over and returns the organism to its normal state.

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Symptoms of fear in combat flying *Based on reports of combat pilots during World War II.* (From L. F. Shafer (1947) 'Symptoms of Fear in Combat Flying', in *Journal of Consulting Psychology*, 11:137–143.)

During combat missions did you feel...?	*Sometimes*	*Often*	*Total*
A pounding heart and rapid pulse	56%	30%	86%
That your muscles were very tense	53	30	83
Easily irritated or angry	58	22	80
Dryness of the throat or mouth	50	30	80
Nervous perspiration or cold sweat	53	26	79
Butterflies in the stomach	53	23	76
A sense of unreality–that this could not be happening to you	49	20	69
A need to urinate frequently	40	25	65
Trembling	53	11	64
Confused or rattled	50	3	53
Weak or faint	37	4	41
That right after a mission you were unable to remember the details of what had happened	34	5	39
Sick to the stomach	33	5	38
Unable to concentrate	32	3	35
That you had wet or soiled your pants	4	1	5

requiring substantial physical energy (such as attack or flee; the role of this fight-or-flight response in threatening or stressful situations

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Undoing effect of Positive Emotions

- Participants were first asked to prepare a speech on 'Why you are a good friend' under considerable time pressure.
- They were told that the speech would be videotaped and evaluated by their peers.
- This speech task produced feelings of anxiety, along with increases in blood pressure, heart rate, and other indices of cardiovascular activity.
- These physiological changes lingered on, even after participants were told that they would not have to deliver their speech after all.
- At this point, the participants were shown a randomly selected film clip that induced one of two positive emotions (joy or contentment), a negative emotion (sadness), or no emotion.
- Cultivating positive emotions, then, appears to be a particularly good way to combat the lingering physiological after effects of negative emotions.



Figure 11.5 The Undoing Effect of Positive Emotions. *Joy and contentment produce faster cardiovascular recovery from lingering negative emotional arousal than neutrality and sadness.* (Adapted with permission from B. L. Fredrickson, R. A. Mancuso, C. Branigan, & M. M. Tugade (2000). 'The Undoing Effect of Positive Emotions', in Motivation and Emotion, 24:237–258. © 2000 Kluwer Academic/Plenum Publishers.)

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